

Continuation on what materials are viable for the rocket

Material	Density (Kg/m ³)	Youngs Modulus (GPa)	Tensile strength (Mpa)	Yield Strength (Mpa)	Elongation at break (%)	Cost (USD/Kg)	3d printability
PET	1350	9	150	40	32	1.52	Very good
HDPE	950	0.8	28	33	367	0.93	Good
LDPE	920	0.3	20	10	200	1.19	Do-able
LDF	350				0	1.94	No but can be machined
MDF	700	4	18		0		No but can be machined
HDF	900	10.5	35	25	0		No but can be machined

Material evaluation

PET (Polyethylene terephthalate)

PET, which is the plastic bottles are normally made of, is a material that is denser than most of the other plastics available. Its 3d printable which means that it can be sold as an STL file, but its tensile strength is much higher than the other materials we have available which would allow it to hold up better than most other options. PET is also very cheap option meaning it could be suitable for the task.

Other materials

While looking at the other materials for some such as LDF enough data couldn't be gathered to be measured and the price of MDF and HDF can't be calculated based on per Kg however based on the price of the boards it would be quite expensive. The main thing which makes the other materials unsuitable however is the low tensile strength compared to our other options. Tensile strength is important for the rocket as it is what will allow it to hold up while the rocket is moving. On top of this while looking for common materials that couldn't be 3d printed the manufacturing process for the materials that would need to be machined would take too long and add a sizable amount of money for the consumer at the end.

